

Forecast-to-Execution Interfaces in Forecasting-Driven Systems: Why Evaluation Must Follow Realized Actions Under Frictions

Abstract

Forecasting-driven systems are often evaluated as if proposed outputs were immediately realized. Once constrained downstream implementation layers intervene under frictions, however, proposed targets and realized actions can diverge. Forecast quality and realized system quality therefore become distinct evaluation objects. We study a forecast-to-execution interface that separates proposed targets from realized executed decisions and attaches primary evaluation to the executed path while holding predictive information fixed. In a narrow cost-sensitive portfolio case study with a frozen learned policy, the validation-selected interior operating point $\eta = 0.5$ improves held-out executed-path net Sharpe relative to $\eta = 1.0$ in positive-cost regimes while roughly halving executed turnover, and the zero-cost row remains near-flat. Accounting diagnostics and conservative supporting analyses support the same conclusion: this is an implementation-side evaluation result rather than evidence of a stronger predictor.

1 Introduction

Forecasting-driven systems are often evaluated at the level of proposed outputs, as if those outputs were directly realized by downstream decision layers. Under frictions or implementation constraints, however, proposed targets need not coincide with realized actions. This creates an evaluation problem: forecast quality and realized decision quality are not the same object once a forecast-to-decision interface intervenes. The central question here is not whether a stronger predictor has been found, but how a forecasting-driven system should be evaluated once proposed outputs are not directly realized.

We study this general evaluation issue in a concrete cost-sensitive portfolio case study. The portfolio setting is the evidence source rather than the paper’s identity: predictive information is held fixed, target-level quantities remain diagnostic only, and primary evaluation is attached to executed-path outcomes as the forecast-to-execution interface is varied. In the central selected-point comparison, the validation-selected interior operating point $\eta = 0.5$ is evaluated against the immediate-execution baseline $\eta = 1.0$; the positive-cost rows improve, while the zero-cost row remains near-flat. We submit this paper to a forecasting workshop not because of the financial application alone, but because it isolates a forecasting-system evaluation question: once predictive outputs are mediated by constrained downstream implementation layers, forecast-side quality and realized system quality become distinct objects.

The paper makes three narrow contributions. First, it contributes an evaluation principle for forecasting-driven systems under frictions: once proposed targets and realized actions diverge, executed-path evaluation should be primary while target-level quantities remain diagnostic. Second, it contributes an identification design that holds predictive information fixed and varies only the downstream forecast-to-execution interface in a frozen-policy comparison. Third, it contributes evidence for that evaluation reading from the selected-point result, the accounting diagnostics, and a conservative support package that includes cost sweep recurrence, fixed-universe recurrence, decision-architecture recurrence, and the master target-versus-executed audit.

2 Forecasting-System Evaluation Under Downstream Constraints

Figure 1 gives the paper’s core framing. Forecast outputs first become proposed targets and only later become realized actions after passing through an execution interface. Once frictions or implementation constraints intervene, the proposed target is no longer the same object as the realized decision. That distinction matters because the system is ultimately experienced through realized trading, realized cost, and realized decision quality rather than through the counterfactual target alone.

The interface perspective changes the evaluation question. The issue is not whether target-level quantities are meaningless; they remain useful diagnostics. The issue is that they answer a different question from the one the deployed decision system actually faces. For that reason, this workshop paper treats the executed path as the primary evaluation object and uses target-level summaries only to explain where target and realized paths diverge.

3 Experimental Setup

The empirical setting is intentionally narrow. We use a fixed 27-name portfolio snapshot under the canonical frozen split and hold the learned policy and predictive information fixed across execution arms. The main comparison is therefore not a retraining study. Instead, it isolates how realized outcomes change when only the forecast-to-execution interface is varied.

The selected-point protocol compares the validation-selected interior operating point $\eta = 0.5$ against the immediate-execution baseline $\eta = 1.0$ on the main cost rows $\kappa \in \{0, 5 \times 10^{-4}, 10^{-3}\}$. The primary metric is executed-path net Sharpe. Executed turnover is reported as supporting evidence because the mechanism of improvement is expected to operate through realized trading intensity and realized cost. This setup keeps the paper focused on one identification question: whether changing only the forecast-to-execution interface changes realized decision quality under frictions.

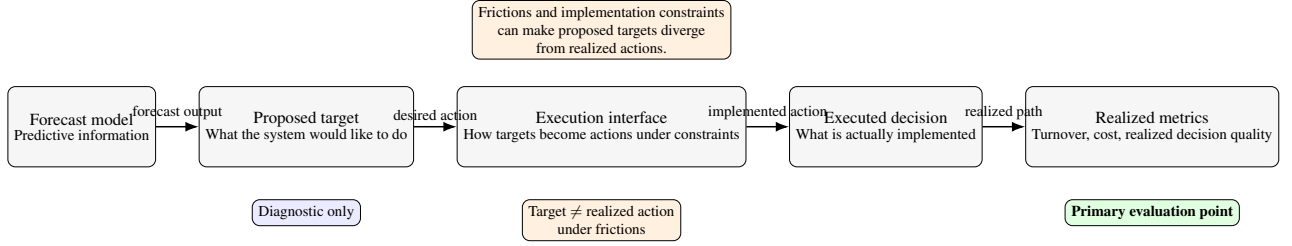


Figure 1: From forecasts to realized decisions under frictions. The conceptual diagram shows how predictive outputs become proposed targets and then realized actions once a forecast-to-execution interface and frictions intervene. Its role is to make the workshop framing explicit: forecast quality and realized decision quality are related but not identical, so evaluation should follow realized actions rather than targets alone. Its role is to frame the paper as an evaluation problem in forecasting-driven systems rather than as a new execution-policy proposal.

κ	Net Sharpe ($\eta = 1.0$)	Net Sharpe ($\eta = 0.5$)	Paired-median Δ Sharpe	TOexec ($\eta = 1.0$)	TOexec ($\eta = 0.5$)
0	1.1542	1.1554	-0.0002	0.02200	0.01095
5×10^{-4}	1.1340	1.1447	+0.0105	0.02200	0.01095
10^{-3}	1.1127	1.1340	+0.0213	0.02200	0.01095

Table 1: Held-out executed-path comparison for the frozen-policy selected operating point. With predictive information held fixed, the validation-selected interior point $\eta = 0.5$ is compared against the immediate-execution baseline $\eta = 1.0$ on $\kappa \in \{0, 5 \times 10^{-4}, 10^{-3}\}$, showing positive-cost gains with a large reduction in executed turnover while the zero-cost row remains near-flat. The paired-median delta is computed on aligned seed-level outcomes and therefore need not equal the arithmetic difference between the displayed marginal medians. This is the paper’s only main empirical result, because it isolates how the forecast-to-execution interface changes realized decision quality under frictions.

4 Results

The first empirical question is evaluative rather than domain-specific: once frictions separate proposed targets from realized actions, does changing only the forecast-to-execution interface change realized system quality? Table 1 answers that question in the paper’s concrete portfolio case study. With the learned policy and predictive information held fixed, the validation-selected interior point $\eta = 0.5$ improves held-out executed-path net Sharpe relative to the immediate-execution baseline $\eta = 1.0$ by $+0.0105$ at $\kappa = 5 \times 10^{-4}$ and $+0.0213$ at $\kappa = 10^{-3}$, while reducing median executed turnover from 0.02200 to 0.01095. The $\kappa = 0$ row remains near-flat at -0.0002 , which is the expected zero-cost pattern under the documented interpretation. This is therefore best read as an implementation-side result that appears under frictions rather than as evidence of a stronger predictor.

The key point is not merely that $\eta = 0.5$ is better in the positive-cost rows. The key point is that target-based and executed-based evaluation answer different questions once frictions intervene.

Figure 2 explains why executed-path evaluation is primary. Target-level quantities are diagnostic, not headline evaluation objects. At the validation-selected operating point $\eta = 0.5$, target turnover is about twice realized executed turnover across the main cost rows, while tracking remains small but nonzero and the final path gap widens with cost. These diagnostics matter because proposed targets and realized positions are not the same object once frictions intervene. A narrow target-versus-executed evaluation comparison reaches the same conclusion more directly: in the positive-cost rows, executed-path evaluation favors $\eta = 0.5$, while

target-based evaluation is approximately flat to slightly negative. In other words, the choice of evaluation object is not merely presentational: in the positive-cost rows, target-based evaluation weakens or erases the selected-point improvement that remains visible on the realized executed path. The accounting package therefore explains the evaluation object used in the main result rather than competing with it.

A denser cost sweep in the appendix remains directionally compatible with the same friction-sensitive reading: the zero-cost row stays near-flat, while the realized executed-path advantage remains positive across the positive-cost rows.

The point is not that smoothing is intrinsically beneficial; rather, once frictions separate desired targets from realized actions, the forecast-to-execution interface becomes part of the evaluated system, and its effects must be assessed on the realized path.

A conservative supporting analysis, reported in the appendix under the title *Similar forecasting information, different realized decision quality*, shows that forecast-side metrics move only slightly while decision-side metrics move much more strongly in the positive-cost rows. At $\kappa = 5 \times 10^{-4}$ and $\kappa = 10^{-3}$, forecast MSE shifts by only -0.03% and $+0.28\%$, whereas executed turnover and realized cost fall by 90.8% and 95.1%, with net Sharpe rising by $+0.9539$ and $+2.0208$. These support-package deltas are not directly comparable in scale to the selected-point RL deltas in Table 1, because they come from a different auxiliary comparison setting and are included only to illustrate the asymmetry between small forecast-side movement and large decision-side consequences. This pattern is consistent with an interface-side interpretation, but because only metric-level similarity is established rather than raw forecast-vector identity, it remains appendix support rather than a main-text finding.

The remaining appendix materials organize the same narrow support package more broadly: fixed-universe recurrence, decision-architecture recurrence, a master target-versus-executed disagreement audit, an appendix-only toy example, and compact meta figures. Taken together, these appendix materials reinforce the same narrow evaluation-object reading without changing the main evidence hierarchy.

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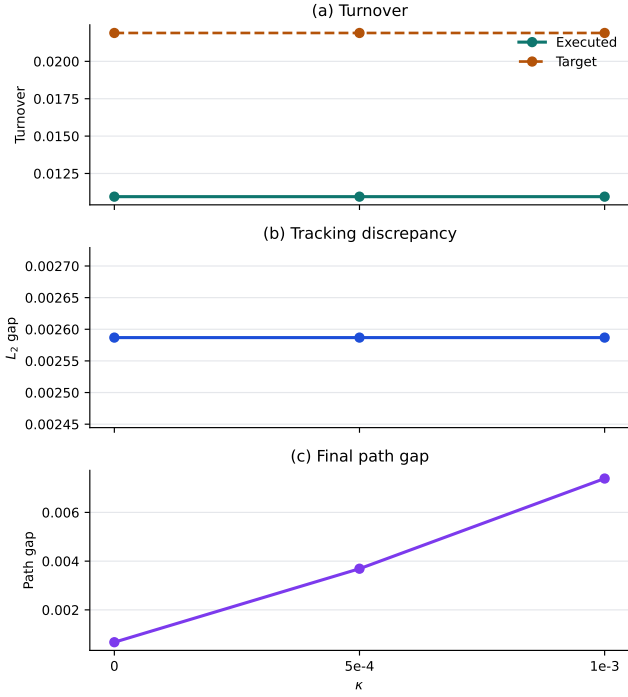


Figure 2: Diagnostic evidence that target and executed paths diverge under frictions. Diagnostic accounting summaries compare target-level and executed-path quantities at the validation-selected operating point $\eta = 0.5$. Target turnover remains about twice executed turnover, tracking is small but nonzero, and the final path gap widens with cost, showing why the realized executed path rather than the target path is the primary evaluation object in this paper.

5 Discussion and Limitations

This workshop paper makes one narrow point: when frictions separate proposed targets from realized actions, forecasting systems should be evaluated on the realized path they actually induce. In our case study, the selected-point result and the accounting diagnostic support the same interpretation. Holding predictive information fixed, changing the forecast-to-execution interface changes realized turnover, realized cost, and realized decision quality. That is the paper’s contribution, and it is best read as an evaluation result about forecasting-to-decision systems under frictions.

The first limitation is scope. The main result is still a narrow portfolio-domain case study built on a fixed 27-name snapshot under the canonical frozen split. The added appendix support reduces the most local artifact concerns, but it does not replace the main evidence hierarchy and it does not convert the paper into a broad cross-domain study. The paper therefore supports a concrete forecasting-to-decision evaluation argument rather than a universality claim.

The second limitation is the learning setup. The central identification result is a frozen-policy comparison rather than an end-to-end retraining study. That is useful for the workshop question because it keeps predictive information fixed, but it also means the evidence is about how realized decision quality changes under a different interface, not about how a jointly retrained system would behave. The remaining materials should stay in supporting roles: the denser cost-grid figure is appendix support, the cross-universe and cross-architecture

packages remain support-only recurrence checks, and the *Similar forecasting information, different realized decision quality* table should remain conservative because exact forecast identity across arms has not been directly established.

The compact temporal multi-split pilot was mixed, so we do not use it to claim broad temporal robustness in this workshop version. The stronger added support instead comes from fixed-universe recurrence and repeated target-versus-executed disagreement across the denser cost sweep and the execution-layer package, summarized jointly by the master audit. The appendix-only toy example is included only as an intuition aid showing that the same proposal-versus-realization distinction can arise in a generic decision process; it is not presented as new empirical evidence.

These limitations do not weaken the core reading; they locate it. The forecasting relevance of the paper is that it distinguishes forecast quality from realized decision quality in settings where forecasts pass through downstream implementation layers before becoming realized actions. A natural next step is to test the same evaluation logic in additional forecasting-driven domains, but the workshop contribution already stands on the narrower point established here: under frictions, evaluation should follow realized actions rather than targets alone.

6 Conclusion

This paper argues for a simple but important evaluation principle for forecasting-driven decision systems under frictions: realized actions, not proposed targets alone, should anchor primary evaluation. In the concrete portfolio case study here, the validation-selected interior execution interface improves positive-cost held-out realized performance while the zero-cost row remains near-flat. The added appendix support suggests that, under positive frictions, executed-path evaluation can matter materially and that this is not merely a single U27 accounting curiosity within the documented support settings. That support remains secondary, however, and we do not use it to claim broad temporal robustness or universal generalization beyond the settings documented here.

Universe	κ	Exec _{1.0}	Exec _{0.5}	Δ Exec	Tgt _{1.0}	Tgt _{0.5}	Δ Tgt	TO red.%	Disag.	Near-flat@0	Pos-cost dir.	Verdict
Current	0	1.1789	1.1783	0.0010	1.1789	1.1789	0.0000	48.9	no	yes	n/a	Green
Current	5e-4	1.1575	1.1687	0.0119	1.1575	1.1573	-0.0003	48.9	yes	n/a	yes	Green
Current	1e-3	1.1354	1.1582	0.0223	1.1354	1.1349	-0.0006	48.9	yes	n/a	yes	Green
Alt-LargeCap	0	0.9841	0.9895	0.0054	0.9841	0.9841	-0.0000	49.7	yes	no	n/a	Yellow
Alt-LargeCap	5e-4	0.9665	0.9807	0.0142	0.9665	0.9666	-0.0001	49.7	yes	n/a	yes	Green
Alt-LargeCap	1e-3	0.9489	0.9719	0.0230	0.9489	0.9490	-0.0002	49.7	yes	n/a	yes	Green
Sector-Balanced	0	1.5914	1.5944	0.0028	1.5914	1.5914	0.0000	48.7	no	yes	n/a	Green
Sector-Balanced	5e-4	1.5708	1.5837	0.0110	1.5708	1.5700	-0.0006	48.7	yes	n/a	yes	Green
Sector-Balanced	1e-3	1.5502	1.5730	0.0192	1.5502	1.5486	-0.0011	48.7	yes	n/a	yes	Green

Universe-level summary: Current = Green, Sector-Balanced = Green, Alt-LargeCap = Yellow (mixed only in the zero-cost row).

Table A1: Held-out multi-universe support summary for the locked ‘eta=1.0’ versus ‘eta=0.5’ comparison. Marginal Sharpe and turnover columns report seed medians; the delta columns report paired seed-aligned medians. ‘Disag.’ marks target-versus-executed disagreement, ‘Near-flat@0’ marks the documented zero-cost near-flat check, and ‘Pos-cost dir.’ marks positive-cost executed-path direction.

A. Cost Sweep Support. The appendix cost sweep keeps the same evaluation-object reading on a denser six-point grid. It remains support-only and should not be promoted into a second main result.

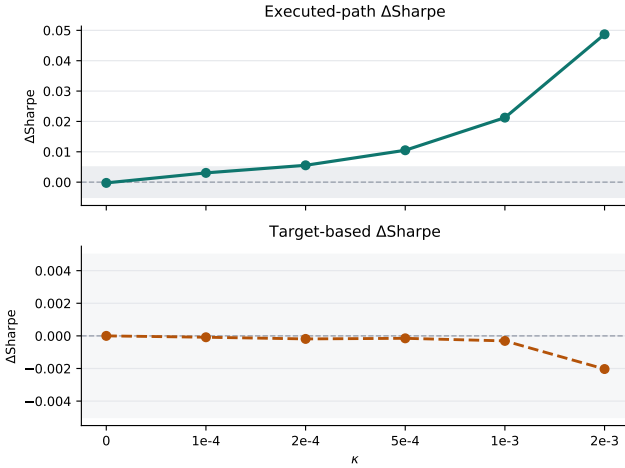


Figure A1: Denser cost-grid support for the friction-sensitive evaluation pattern. The executed-path delta remains near-flat at $\kappa = 0$ and stays positive across the positive-cost rows, while the target-based delta remains approximately zero or negative. The two panels are shown on separate scales; the visual point is directional contrast, not equal magnitude. This is appendix support for the paper’s evaluation-object reading, not a new main result class.

B. Multi-Universe Robustness. The fixed-universe package asks whether the same evaluation-object discrepancy remains visible outside the exact current U27 basket. In the current summary, ‘Current’ and ‘Sector-Balanced’ remain Green, while ‘Alt-LargeCap’ is mixed because its zero-cost row is slightly above the documented near-flat threshold even though its positive-cost rows remain directionally compatible.

C. Decision-Architecture Robustness. The architecture package remains support-only. Beyond the main RL row, two independent non-RL families are now included: a deadband family and a volatility-scaled family. Each is shown with champion and runner-up configurations selected conservatively, and both families contribute additional positive-cost ‘ranking mismatch’ rows. The linear/prox family remains mixed because target-based and executed-based evaluation stay numerically aligned there.

Architecture	κ	Exec(tr)	Exec(ed)	Δ Exec	Tgt(tr)	Tgt(ed)	Δ Tgt	TO red.%	Disag.	Near-flat@0	Pos-cost dir.	Support verdict
RL-Selected	0	1.1587	1.1616	0.0029	1.1587	1.1587	-0.0000	48.9	none	yes	n/a	Green
RL-Selected	5e-4	1.1388	1.1514	0.0126	1.1388	1.1384	-0.0005	48.9	ranking_mismatch	n/a	yes	Green
RL-Selected	1e-3	1.1190	1.1413	0.0223	1.1190	1.1181	-0.0009	48.9	ranking_mismatch	n/a	yes	Green
Deadband-Champion	0	-5.9405	-5.9227	0.0078	-5.9405	-5.9405	0.0000	0.2	none	yes	n/a	Green
Deadband-Champion	5e-4	-6.1148	-6.1068	0.0080	-6.1148	-6.1149	-0.0001	0.2	ranking_mismatch	n/a	yes	Green
Deadband-Champion	1e-3	-6.2883	-6.2801	0.0082	-6.2883	-6.2886	-0.0003	0.2	ranking_mismatch	n/a	yes	Green
VolScaledFia-Champion	0	-5.9405	-5.9229	0.0076	-5.9405	-5.9405	0.0000	0.3	none	yes	n/a	Green
VolScaledFia-Champion	5e-4	-6.1148	-6.1105	0.0042	-6.1148	-6.1148	-0.0000	0.3	ranking_mismatch	n/a	yes	Green
VolScaledFia-Champion	1e-3	-6.2883	-6.2844	0.0039	-6.2883	-6.2883	-0.0000	0.3	ranking_mismatch	n/a	yes	Green
Rule-EtaFixed	0	1.1587	1.1616	0.0029	1.1587	1.1587	-0.0000	48.9	none	yes	n/a	Green
Rule-EtaFixed	5e-4	1.1388	1.1514	0.0126	1.1388	1.1384	-0.0005	48.9	ranking_mismatch	n/a	yes	Green
Rule-EtaFixed	1e-3	1.1190	1.1413	0.0223	1.1190	1.1181	-0.0009	48.9	ranking_mismatch	n/a	yes	Green
Linear-Prox	0	1.2129	1.0074	-0.2055	1.2129	1.0074	-0.2055	88.4	none	no	n/a	Red
Linear-Prox	5e-4	0.0923	0.8910	0.7987	0.0923	0.8910	0.7987	88.4	none	n/a	yes	Yellow
Linear-Prox	1e-3	-1.0260	0.7746	1.8005	-1.0260	0.7746	1.8005	88.4	none	n/a	yes	Yellow

TO red.% is computed relative to the reference arm within each architecture family. Small values for the deadband and volatility-scaled families reflect near-matched turnover within those family-local comparisons and should not be read as cross-family inefficiency. Support verdicts are also family-local; runner-up configurations remain in the CSV and notes and are omitted from this compact visible table only for readability.

Table A2: Support-only decision-architecture summary under frozen-source comparisons. Verdicts are computed within each architecture family using the documented zero-cost, positive-cost, and disagreement checks; the table is not intended for cross-family ranking by absolute Sharpe level. ‘Disag.’ reports the conservative target-versus-executed disagreement label, ‘Near-flat@0’ reports the documented zero-cost near-flat check, and ‘Pos-cost dir.’ reports positive-cost executed-path direction.

D. Target-versus-Executed Master Audit. The master audit pools cost rows, fixed universes, architecture families, and the optional canonical split reference into one conservative disagreement map. The aggregate remains dominated by ‘ranking mismatch’ rows, with a smaller number of ‘sign flip’ and ‘magnitude only’ cases, reinforcing the same narrow point: under positive frictions, the evaluation object can change the interpretation.

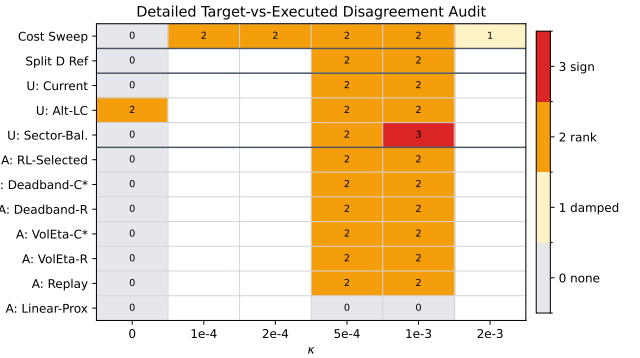


Figure A2: Master target-versus-executed disagreement audit. Group separators distinguish cost rows, the canonical split reference, fixed universes, and architecture families; champion configurations are marked explicitly. The map keeps the wording narrow: repeated disagreement is visible, but not every auxiliary architecture produces the same pattern.

E. Minimal Domain-General Toy Example. The toy example is appendix-only and illustrative rather than empirical evidence. Its role is only to show that a proposal-versus-realization mismatch can create a target-versus-executed evaluation gap in a generic one-dimensional decision process. The zero-friction row remains in near-agreement, a very low-friction row remains effectively tied, and higher-friction rows turn into ‘ranking mismatch’.

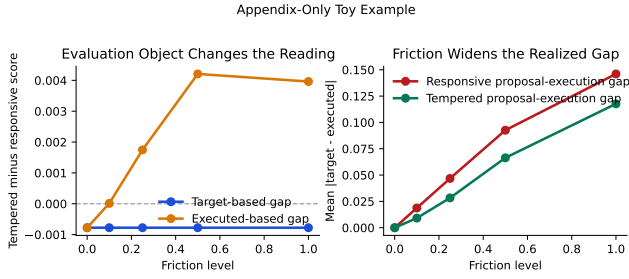


Figure A3: Appendix-only toy example. The left panel shows that target-based and executed-based evaluation can diverge as friction increases even in a generic scalar decision process; the right panel shows the growing proposal-versus-executed gap that makes the mismatch plausible. This figure is illustrative only and is not empirical evidence from the portfolio setup.

F. Meta Figures. The compact meta figures keep the emphasis on fixed-universe recurrence and execution-layer recurrence rather than on the compact temporal pilot. The temporal pilot was mixed and is acknowledged that way: it is not the main visual focus of this appendix package.

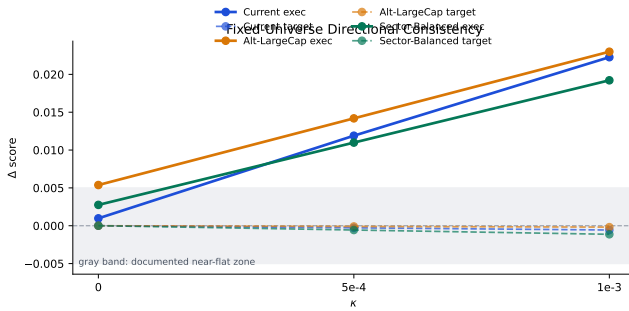


Figure A4: Meta-level fixed-universe consistency summary. The figure emphasizes directional compatibility rather than cross-universe equality of absolute gains, and it keeps the emphasis on the tested fixed universes rather than on a broader temporal robustness claim.

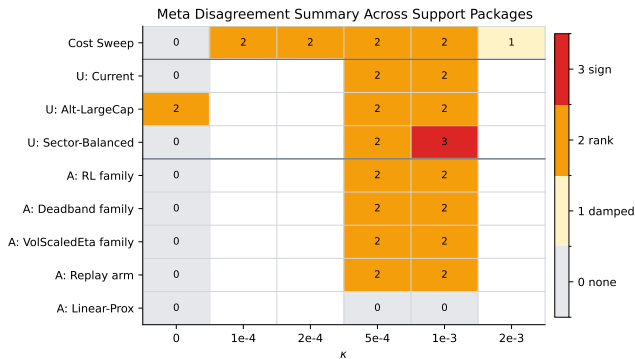


Figure A5: Meta disagreement map across the cost sweep, fixed universes, and architecture families. This compact figure is a family-level summary rather than a detailed audit. The compact temporal pilot is not the visual center here because that pilot was mixed; the stronger support in this round comes from fixed-universe and execution-layer recurrence.

G. Similar Forecasting Information, Different Realized Decision Quality.

This table is included only as conservative support for the interface-side interpretation and comes from an auxiliary comparison setting rather than from the main selected-point RL result.

κ	$\Delta\text{MSE}(\%)$	$\Delta\text{Sign Acc. (pp)}$	$\Delta\text{Rank IC}$	Relative TOexec reduction (%)	Relative realized-cost reduction (%)	$\Delta\text{Net Sharpe}$
0	0.00	0.00	0.0000	0.0	0.0	0.0000
5×10^{-4}	-0.03	+0.41	+0.0038	90.8	90.8	+0.9539
10^{-3}	+0.28	+0.49	+0.0033	95.1	95.1	+2.0208

Table A3: Forecast-side movement remains small while decision-side movement is large in the positive-cost rows of the auxiliary support package. These deltas are not intended to be compared numerically with the selected-point RL deltas in Table 1. Only metric-level similarity is established here, so the table does not claim exact forecast identity.